

Python & SQL Alchemy Republish Script Task

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Task Background:

We track historical data for all products on Walmart.

BOOM – we lost some data and need to restore it using API (What's API? Google it.)

Task Overview:

Yo, what's up? In this task, we are going to perform a “republish” script.

This script is gonna “fill” missing data by using the provided API.

Unfortunately, the data we can obtain from the API is not all the data we need to save, but it's something – so let's save as much as we can.

We are going to fetch the data and fill the other parts with default values. We'll get to that later. First of all, see some junk code and info about what you are going to use.

API Information (for fetching historical data):

API docs for fetching historical data: <https://dataspark.co/api/docs/>

ApiKey: 717d12a2ffbfa6752e91ae36f9687284

WARNING: Do not overload the API with excessive usage, avoid running infinite loops, and use it wisely.

SQL Alchemy (What's that? Google it.) Table for Historical Data – Python Code:

```
from sqlalchemy import Column, DateTime, Float, ForeignKey, Integer, JSON, String, func
from .base import Base
```

```
class MarterTrackerProductHistory(Base):
    __tablename__ = 'MarterTrackerProductHistory'

    id = Column(Integer, primary_key=True, autoincrement=True)
```

```
date = Column(DateTime, nullable=False, default=func.now(), index=True)
item_id = Column(String, ForeignKey('MarterTrackerProduct.item_id'), index=True)
review_count = Column(Integer)
rating = Column(Float)
offers = Column(String)
```

```
current_seller = Column(JSON)
""" JSON Structure:
# seller_id = Column(String)
# seller_display_name = Column(String)
# seller_avg_rating = Column(Float)
# seller_review_count = Column(Integer)
# seller_wfs_enabled = Column(Boolean, default=False)
"""
```

```
current_stock = Column(JSON)
""" JSON Structure:
# stock = Column(Integer, nullable=False, default=0)
# stock_wfs = Column(Integer, default=0)
# stock_sf = Column(Integer, default=0)
# stock_walmart = Column(Integer, default=0)
"""
```

```
current_price = Column(JSON)
""" JSON Structure:
# price = Column(Float, nullable=False)
# price_lowest = Column(Float)
# price_walmart = Column(Float)
# price_lowest_wfs = Column(Float)
# price_lowest_sf = Column(Float)
"""
```

```
current_offers = Column(JSON)
""" JSON Structure:
# offer_count = Column(Integer)
# offer_count_wfs_sellers = Column(Integer)
# offer_count_sf_sellers = Column(Integer)
"""
```

```
current_badges = Column(JSON)
""" JSON Structure:
```

```

# badge_best_seller = Column(Boolean, default=False)
# badge_popular_pick = Column(Boolean, default=False)
# badge_x_amount_sold = Column(Integer)
# badge_x_amount_in_cart = Column(Integer)
"""

```

Record Structure Example:

```

{
  "date": datetime.now(timezone.utc),
  "review_count": data["reviews"],
  "rating": data["rating"],
  "offers": json.dumps(all_offers),
  "current_seller": {
    "sellerId": data["sellerId"],
    "seller_display_name": data["sellerDisplayName"],
    "seller_avg_rating": data["sellerAverageRating"],
    "seller_review_count": data["sellerReviewCount"],
    "seller_wfs_enabled": data["wfsEnabled"],
  },
  "current_stock": {
    "stock": stock,
    "stock_wfs": stock_wfs,
    "stock_sf": stock_sf,
    "stock_walmart": stock_walmart,
  },
  "current_price": {
    "price":
data["price"],
    "price_lowest": price_lowest,
    "price_lowest_wfs": price_lowest_wfs,
    "price_lowest_sf": price_lowest_sf,
    "price_walmart": price_walmart,
  },
  "current_offers": {
    "offer_count": offer_count,
    "offer_count_wfs_sellers": offer_count_wfs_sellers,
    "offer_count_sf_sellers": offer_count_sf_sellers,
  },
  "current_badges": {

```

```

    "badge_best_seller": data["badge_best_seller"],
    "badge_popular_pick": data["badge_popular_pick"],
    "badge_x_amount_sold": data["badge_x_amount_sold"],
    "badge_x_amount_in_cart": data["badge_x_amount_in_cart"],
  },
}

```

Example of a **FULL** Record:

```

{'item_id': '1170687181', 'id': 1025910992168091649, 'rating': 4.7, 'current_seller':
{'sellerId': 101058484, 'seller_avg_rating': 5, 'seller_display_name': 'S&A Distribution',
'seller_review_count': 4, 'seller_wfs_enabled': True}, 'current_price': {'price': 21.99,
'price_lowest': 21.99, 'price_lowest_sf': 22.95, 'price_lowest_wfs': 21.99,
'price_walmart': 24.97}, 'current_badges': {'badge_best_seller': False,
'badge_popular_pick': False, 'badge_x_amount_in_cart': 0, 'badge_x_amount_sold': 0},
'review_count': 2233, 'date': datetime.datetime(2024, 12, 2, 15, 33, 6, 958782), 'offers':
['{"sellerId": 101058484, "sellerName": "S&A Distribution", "price": 21.99, "stock": 188,
"wfs": true}, {"sellerId": 0, "sellerName": "Walmart.com", "price": 24.97, "stock": 11780,
"wfs": true}, {"sellerId": 13597, "sellerName": "BEYOND EVERYTHING LLC", "price": 24.8,
"stock": 1, "wfs": true}, {"sellerId": 101175862, "sellerName": "Agents of S.A.L.E.S.",
"price": 24.9, "stock": 11, "wfs": true}, {"sellerId": 101661420, "sellerName": "Prime
Essentials Inc.", "price": 22.95, "stock": 5, "wfs": false}, {"sellerId": 101087433,
"sellerName": "Industrial Resolution", "price": 24.75, "stock": 10, "wfs": false},
{"sellerId": 101552809, "sellerName": "TH&Comapny", "price": 24.99, "stock": 8, "wfs":
false}, {"sellerId": 101014953, "sellerName": "Paramount", "price": 24.95, "stock": 40,
"wfs": false}, {"sellerId": 1145, "sellerName": "Toywiz", "price": 26.88, "stock": 6, "wfs":
false}, {"sellerId": 101434374, "sellerName": "trendy cool store llc", "price": 28.49,
"stock": 1, "wfs": false}, {"sellerId": 101040129, "sellerName": "Love Happy Customers",
"price": 29.99, "stock": 4, "wfs": true}, {"sellerId": 101469047, "sellerName": "Larson-
Bros", "price": 30, "stock": 3, "wfs": true}, {"sellerId": 101026588, "sellerName": "We-R-
Toys", "price": 29.95, "stock": 161, "wfs": false}, {"sellerId": 101415957, "sellerName":
"Tekversify Shop", "price": 31.48, "stock": 3, "wfs": false}, {"sellerId": 101501926,
"sellerName": "MRI Trading LLC", "price": 31.86, "stock": 3, "wfs": false}, {"sellerId":
12988, "sellerName": "DropAir", "price": 32.82, "stock": 30, "wfs": false}, {"sellerId":
101319747, "sellerName": "AMZN ENTERPRISES LLC", "price": 33.99, "stock": 12, "wfs":
true}, {"sellerId": 101502408, "sellerName": "Tittlers Ecom LLC", "price": 33.72, "stock":
20, "wfs": false}, {"sellerId": 101035778, "sellerName": "JBay Treasures", "price": 34.99,
"stock": 2, "wfs": true}, {"sellerId": 101113599, "sellerName": "Parker Central LLC",
"price": 34.99, "stock": 1, "wfs": true}, {"sellerId": 14451, "sellerName": "CaveGamers",
"price": 34.79, "stock": 10, "wfs": false}, {"sellerId": 101438862, "sellerName": "Zillion

```

Tech & Trade LLC", "price": 28, "stock": 2, "wfs": false}, {"sellerId": 101402899, "sellerName": "Market Place of USA", "price": 34.99, "stock": 69, "wfs": false}, {"sellerId": 101530022, "sellerName": "7NOV7 INC.", "price": 34.99, "stock": 5, "wfs": false}, {"sellerId": 101523155, "sellerName": "ejlb llc", "price": 35, "stock": 1, "wfs": false}, {"sellerId": 101196285, "sellerName": "CLICK HUNTER CORP", "price": 35.6, "stock": 100, "wfs": false}, {"sellerId": 101554527, "sellerName": "Kory M Wood LLC", "price": 36.99, "stock": 11, "wfs": false}, {"sellerId": 101044103, "sellerName": "METROGOODS USA", "price": 38.09, "stock": 60, "wfs": false}, {"sellerId": 101020805, "sellerName": "FUNTOYS", "price": 39, "stock": 1, "wfs": false}, {"sellerId": 101500641, "sellerName": "WORKTOOLS", "price": 39.05, "stock": 50, "wfs": false}, {"sellerId": 3107, "sellerName": "Dragon\'s Gold", "price": 38.88, "stock": 12, "wfs": false}, {"sellerId": 101063380, "sellerName": "Family Supply Co", "price": 39.17, "stock": 17, "wfs": false}, {"sellerId": 101179004, "sellerName": "SGO LINE INC", "price": 38.99, "stock": 5, "wfs": false}, {"sellerId": 101500645, "sellerName": "CCHPRODUCT1985", "price": 39, "stock": 15, "wfs": false}, {"sellerId": 1985, "sellerName": "Collectibles ONLY", "price": 39.94, "stock": 1, "wfs": true}, {"sellerId": 101352613, "sellerName": "SKC Brands LLC", "price": 39.97, "stock": 1, "wfs": true}, {"sellerId": 101288318, "sellerName": "Blossom Row Merch", "price": 40, "stock": 1, "wfs": true}, {"sellerId": 101554319, "sellerName": "Irish Iron Ventures", "price": 39.99, "stock": 10, "wfs": false}, {"sellerId": 101251160, "sellerName": "Double Ach Deals", "price": 39.99, "stock": 6, "wfs": false}, {"sellerId": 101003614, "sellerName": "Mighty Mart", "price": 42, "stock": 36, "wfs": true}, {"sellerId": 101438781, "sellerName": "Clear Vision Goods", "price": 32.05, "stock": 3, "wfs": false}, {"sellerId": 101122848, "sellerName": "Kanazta", "price": 42.18, "stock": 1, "wfs": true}, {"sellerId": 101496735, "sellerName": "CAG Services", "price": 43, "stock": 1, "wfs": false}, {"sellerId": 101053976, "sellerName": "Texas Star", "price": 42.98, "stock": 4, "wfs": false}, {"sellerId": 101067339, "sellerName": "Barbi\'s Bargains", "price": 44.99, "stock": 9, "wfs": true}, {"sellerId": 4676, "sellerName": "Toy Crate, Inc", "price": 44.97, "stock": 3, "wfs": false}, {"sellerId": 101275085, "sellerName": "Sulit Selections", "price": 45, "stock": 1, "wfs": false}, {"sellerId": 101328456, "sellerName": "House Of Goods LLC", "price": 44.99, "stock": 1, "wfs": false}, {"sellerId": 101092496, "sellerName": "Amazing Southern Gifts", "price": 44.99, "stock": 5, "wfs": false}, {"sellerId": 11774, "sellerName": ".1Deals", "price": 44.88, "stock": 33, "wfs": false}, {"sellerId": 101556606, "sellerName": "HUNTER LGCY", "price": 45, "stock": 1, "wfs": false}, {"sellerId": 11567, "sellerName": "Rocky Mountain Textbooks", "price": 44.95, "stock": 1, "wfs": false}, {"sellerId": 101636996, "sellerName": "Go Indigo LLC", "price": 45.77, "stock": 12, "wfs": false}, {"sellerId": 101391736, "sellerName": "Davis Store", "price": 45.9, "stock": 2, "wfs": false}, {"sellerId": 9313, "sellerName": "Teddy Bear Toys inc", "price": 47.99, "stock": 4, "wfs": false}, {"sellerId": 1148, "sellerName": "UnbeatableSale", "price": 49.5, "stock": 20, "wfs": false}, {"sellerId": 101373823, "sellerName": "UnbeatableSale Local", "price": 49.3, "stock": 20, "wfs": false}, {"sellerId": 1812, "sellerName": "Bed Bath & Beyond", "price": 49.87, "stock": 50, "wfs": false}], 'current_stock': {'stock': 12873,

```
'stock_sf': 822, 'stock_walmart': 11780, 'stock_wfs': 12051}, 'current_offers':  
{'offer_count': 58, 'offer_count_sf_sellers': 43, 'offer_count_wfs_sellers': 15}}
```

Okay! Let's start:

Disclaimer: You are not going to push anything to the database but instead imitate the process.

First step: Read the API documentation and understand how to fetch historical pricing (Lowest Price History). Try to retrieve the price history for item ID "**46480251**".

Step 2: Create records for the table **MarterTrackerProductHistory** (just create the objects) for all the price points. Fill in the other values for which data is unavailable with -1, an empty string, an empty JSON, or an empty list – doesn't really matter just fill them with something empty.

You will end up with a lot of records for each item ID, and that's fine. We want to create a record for every price point we have in the history retrieved from the API.

Did you get those records for item ID "46480251"? Awesome! You're ready for step 3.

Step 3: Go to walmart.com, find 10 different item IDs, and run this republish script in a loop for each item ID. Make a list of these 10 item IDs and iterate through them.

At the end of this loop, write a function to push these new records to SQLAlchemy in bulk. Research how to push records in bulk using SQLAlchemy. Again, you don't need to push for real - just write how you would do it.

That's it! You've "republished" the missing historical data. But we're not done yet.

BONUS We need two additional tasks:

- 1) Add historical data about SELLERS (sellers over time).
- 2) Upload the script to the cloud (AWS), because it will eventually run on millions of item IDs, which isn't ideal for a local environment. Want to know why? Ask me.

You can choose to work on either task. I'll guide you through both but pick just one.

Guide for Task 1:

One of the endpoints is https://dataspark.co/api/v1/product/{product_id}/buybox-stats.

This provides the buy box winner (main seller - Google “What does it mean to win the buy box?” for details) over time, their percentages, the last time they won, and the price.

Think about how we can use this information to fill in some of our data (e.g., `current_seller` on specific dates).

Guide for Task 2:

Let’s Dockerize it. Create a Docker image that includes this script so we can deploy it to a VM (EC2 – AWS cloud) and run it there. Your task is to create the image, build it, and run it locally to demonstrate that it works. Once done, we’ll deploy it together!

Let me know once you’re done or have any questions :)